

What Does AI See through Animals' Eyes? Zero-Shot Activity Recognition from Bio-loggers

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Abstract

Biologging with animal-mounted sensors and cameras enables the long-term observation of wildlife in their natural habitats. However, manually annotating the large amounts of egocentric video data they produce is prohibitively costly and requires expert knowledge. To address this challenge, we propose a zero-shot activity recognition system that combines egocentric video with inertial sensor data and leverages a vision language model (VLM) for automated activity annotation. Our framework detects salient moments via anomaly scores, which are computed from multimodal features, and generates dense video captions for those segments. By comparing generated captions with known activity labels using large language models, our system can both assign existing labels and discover previously unseen activities. We validate our method on real-world datasets collected from seabirds and bears, demonstrating that it can generate meaningful activity annotations without any training data or manual supervision. Our results show promise toward scalable, label-efficient wildlife monitoring. Implementation is available at: https://huggingface.co/Bio-sensing/video_captioning_ubicomp_student_challenge.

CCS Concepts

• **Hardware** → **Sensor applications and deployments**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Bio-logging; Vision Language Model; Auto-annotation; Outlier Detection

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1 Introduction

Ubiquitous computing is a technology that uses sensors and computers placed in the environment to continuously collect and use data. In the past, it has mainly been used in everyday human settings, like homes or workplaces. Recently, its application has expanded to domains such as agriculture, healthcare, and wildlife research. These technologies are useful in places where people cannot easily go or when long-term observation is needed.

In this study, we focus on biologging as a new application of ubiquitous computing. Biologging is a method in which small sensor devices are attached to wild animals such as seabirds, fish, or mammals to collect data about their activity and environment, including our earlier work[14]. These devices can record data such as acceleration, depth, GPS, and video over long periods of time, making it possible to observe natural activities and ecology that were previously difficult to capture.

However, to make use of the video data collected through biologging, it is essential to assign activity labels, such as when and what the animal was doing. This annotation process requires too much time and expert knowledge, making manual labeling extremely costly. In some cases, dozens of hours of footage can be collected from a single animal, and the larger the dataset, the more burdensome the annotation becomes.

Due to the time and accuracy limitations of manual annotation, research aimed at automating this process is increasing. For example, Fabian et al. [5] proposed a method that uses vision language model (VLM) to track wild animals and classify their activity in a zero-shot manner. Similarly, Dussert et al. [4] introduced Wild-Match, a framework that generates detailed visual descriptions of camera trap images and identifies animal species by matching these descriptions to a knowledge base, also in a zero-shot manner. However, all of these studies focus on third-person perspectives (e.g., static camera footage). No prior work has explored zero-shot annotation of egocentric videos captured from animal-mounted cameras.

To address this gap, we propose a method that leverages recent advances in vision language model (VLM) to perform zero-shot labeling of animal activities by combining egocentric videos with sensor data, shown in Figure 1. First, our method not only classifies predefined activities but also detects and suggests previously unseen activities. By employing models such as CLIP, we can match activity instances to natural language descriptions, thereby enabling classification even in the absence of predefined labels. As a result, this approach facilitates the automatic construction of an activity catalog from large-scale video datasets without manual annotation. Moreover, because egocentric videos from animals are both scarce and highly domain-specific, accurate interpretation

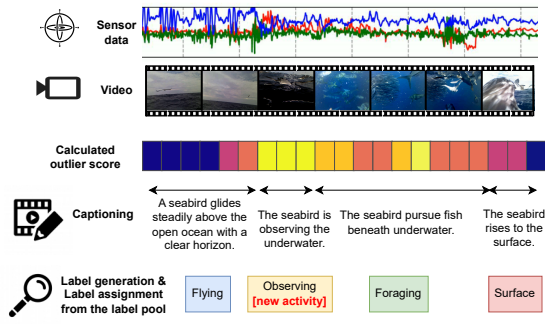


Figure 1: Example of Multi-modal Dense Video Captioning.

ordinarily demands substantial domain expertise. To overcome this limitation, our system labels activities autonomously, removing human intervention; this eliminates personal bias, preserves fairness and consistency, and ultimately provides biologists with reliable, high-quality data.

2 Related Work

2.1 Bio-logging

Bio-logging is a method for studying wild animals by attaching sensors like accelerometers, depth loggers, and cameras to their bodies. It has been used with many species, such as whales, seabirds, seals, and bears [2, 6, 13, 16]. These devices often collect long-term, multimodal data for each individual, sometimes lasting for hundreds of hours.

The Bio-logger Ethogram Benchmark (BEBE) [7] is a large-scale dataset including 1,654 hours of sensor data from 149 animals, annotated with fine-grained activity labels. While this enables activity recognition model development, the study also shows that labeling activities requires expert knowledge and a lot of manual effort.

To make data collection more efficient, Tanigaki et al. [14] proposed an on-board system that detects unusual sensor signals and automatically records short video clips when something unexpected happens. This reduces battery and storage use, and does not require any labeled data. However, it does not provide any explanation of what kind of activity was recorded.

Recent approaches have explored using VLMs like CLIP to recognize activities without supervised training [4, 5]. However, these models are designed for third-person videos where the animal’s body is visible. In contrast, bio-logging cameras typically capture egocentric views, which often lack such visual cues.

To overcome the twin bottlenecks of manual annotation that constrain earlier bio-logging studies, we introduce a multimodal framework that fuses unsupervised sensor outlier detection with egocentric video analysis. IMU-driven frame selection isolates the most informative moments, and a VLM-based auto-annotation assigns activity labels without human intervention.

2.2 Dense Video Captioning

Research on video captioning has advanced from generating a single descriptive sentence for a short clip to handling long, untrimmed

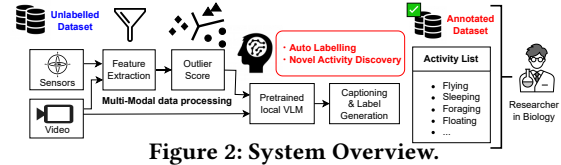


Figure 2: System Overview.

videos that contain multiple events[1, 8]. Because early approaches summarised only the predominant action, they proved inadequate for videos with several distinct activities. This limitation motivated the emergence of *dense video captioning*.

The task was formalised by Krishna et al.[9]. Unlike conventional captioning, dense video captioning jointly (i) detects the temporal boundaries of each event and (ii) produces an individual, semantically rich caption for every detected segment, thereby offering a detailed understanding of the video.

WS-DEC, introduced by Duan et al.[3], was the first framework to treat dense video captioning as a unified problem of event detection and caption generation, explicitly aligning each caption with its corresponding temporal window. Building on this idea, Zhu et al.[19] proposed an end-to-end sequence-generation model that integrates detection and captioning, achieving superior performance through joint optimisation. More recently, the advent of large language models (LLMs) has inspired hybrid systems that incorporate LLMs into dense captioning pipelines, exemplified by Vid2Seq[17].

The principal difference between existing approaches and our target application is camera placement. In our setting, cameras are attached to the animals’ heads or backs, so the recorded video is egocentric, a configuration seldom examined in dense video captioning research. Moreover, whereas conventional captioning systems describe the visual stream, our objective is to recognize the animals’ own actions from the video, which is what the animal sees.

3 Proposed Method

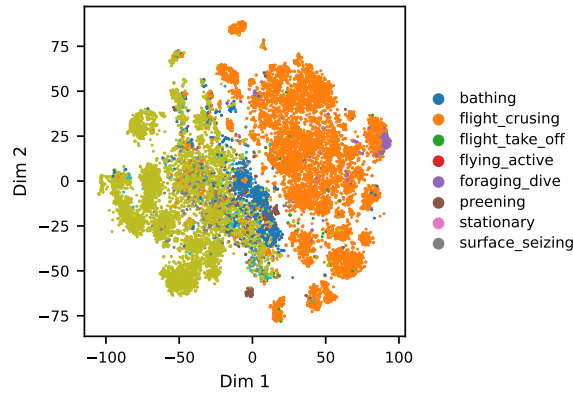
We propose the auto-video captioning system utilizing VLM and Multi-modal data, as illustrated in Figure 2. Our proposed framework consists of **Feature Extraction**, **Outlier Scoring**, **Dense Video Captioning with VLM**, and **Auto Annotation and Dictionary Construction**.

3.1 Feature Extraction

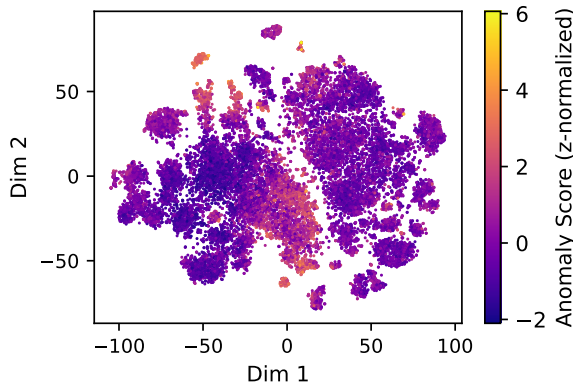
We extract 1-second features from both the accelerometer signals and the video stream. For the sensor modality, we employ the pre-trained encoder from HARNet [18], a foundation model trained on 700,000 person-days of wearable sensor data. Although the encoder is designed for 5-second windows, we feed only the first 1-second portion and use the resulting embedding as our sensor feature representation.

For the visual modality, we use the CLIP vision encoder [11] to compute embeddings from the video frame located at the center of each 1-second interval. This ensures temporal alignment between the two modalities.

Finally, we concatenate the sensor and video feature vectors to form a unified multi-modal representation for each time window. These fused features serve as the basis for anomaly detection and downstream captioning.



(a) Data Visualization with Labels.



(b) Outlier Scoring by Isolation Forest[10].

Figure 3: t-SNE Visualization[15].

3.2 Outlier Scoring

To identify segments that are likely to contain rare or meaningful activities, we compute anomaly scores using the Isolation Forest algorithm. Specifically, the joint multi-modal feature vectors obtained by concatenating the HARNet (sensor) and CLIP (video) embeddings are used as input to the model.

The Isolation Forest[10] is trained in an unsupervised manner over all time windows. It assigns an anomaly score to each window based on the degree of isolation it exhibits in the joint feature space. To make the scores comparable across sessions and individuals, we apply z-score normalization.

These standardized anomaly scores are used to modulate the video frame sampling density in later stages. Windows with higher scores are sampled more densely for captioning, increasing the likelihood of capturing subtle or transient animal activities that would otherwise be overlooked.

The t-SNE visualization[15] in Figure 3 supports our idea, showing the effectiveness of data processing.

3.3 Dense Video Captioning with VLM

After obtaining the sensor-derived scores, we extract video frames and feed them to the VLM. A typical strategy is to sample frames

uniformly throughout the video, but the activities we wish to capture are rare and instantaneous, so uniform sampling may omit the very content that biologists seek. To mitigate this risk, we use the accompanying IMU data to guide frame selection.

By inspecting the anomaly scores, we raise the sampling frequency around segments with high values. Frames located near an outlier, which are detected through an unsupervised learning method, are presumed to capture rare animal activity. Oversampling these segments provides the VLM with richer evidence and consequently improves activity recognition.

We adapted the TimeChat implementation [12], enabling the VLM to receive more appropriately selected frames while incorporating explicit role information into the prompt. In this study, the VLM was instructed to act as a seabird ethologist and segment egocentric video of a streaked shearwater into discrete behavioural events. The details of the prompt and modification can be referred to our implementation.

3.4 Auto Annotation and Dictionary Construction

Once the VLM produces a caption, our system assigns a discrete label to it. The caption remains a free-form natural-language explanation, whereas the label is a single canonical token that designates the activity category.

Recent progress in LLMs enables direct computation of semantic similarity between a caption and a candidate class label written in plain text. Unlike earlier approaches that depend on fixed embedding vectors such as Word2Vec, an LLM can also recognise when a caption does not correspond to any known label. This capacity allows the discovery of novel activities that were not anticipated at design time.

In biologging studies, we begin with an initial dictionary of activity labels before deploying the logger. Even so, the proposed framework continually updates this dictionary by incorporating newly detected activities, thereby enriching the annotation scheme and expanding biological insight.

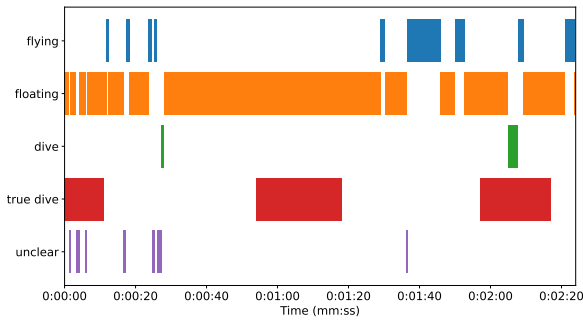
4 Evaluation

4.1 Experimental Settings

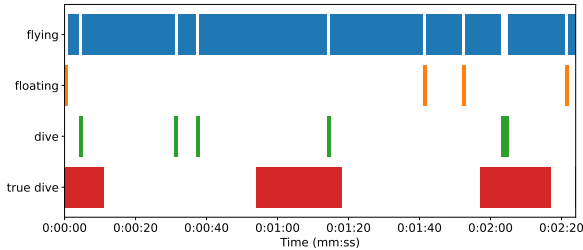
We evaluated our proposed system on a unique biologging dataset originally collected for our previous study[14]. This dataset is composed of recordings from 12 streaked shearwaters (a species of seabird), each equipped with a lightweight biologger recording both egocentric video and tri-axial acceleration data. The data was collected on Awashima Island, Niigata, Japan, with the crucial support of the Yoda Laboratory at Nagoya University. Capturing data in the wild is crucial, as it provides insight into authentic animal behaviors that are difficult to observe in captivity.

Because our system is designed for zero-shot activity recognition, no training data was used. All models—including the CLIP vision encoder and the HARNet sensor encoder—were used in a frozen state without fine-tuning. The Isolation Forest for anomaly scoring was applied in an unsupervised manner to each session individually.

Each 1-second window was represented by a concatenated feature vector from the CLIP and HARNet embeddings. The outlier



(a) Original video (uniform sampling): the first and second *dive* events are entirely missed, and flight segments are mis-classified as *floating* owing to severe motion blur.



(b) Resampled video (proposed): correctly recognises *flying* behaviour and detects all *dive* events; although several segments are over-detected as dives, no true dive is missed.

Figure 4: Labelling results produced by ChatGPT-o3.

scores obtained from Isolation Forest were standardized using z-score normalization within each session. For segments with high anomaly scores, we increased the frame sampling rate to capture more visual information for captioning.

4.2 Result

As an evaluation of our method, we supplied both the original video and its resampled counterpart (described in Section 3) to ChatGPT-o3 and compared the outcomes. The test footage, recorded from a seabird¹, was assigned three initial labels: “flying,” “floating (on the sea),” and “diving into the sea.” In Figure 4, red segments represent diving events, while the other colours denote the labels predicted by ChatGPT-o3. The seabird spends most of the time flying and performs three dives. With the resampled video, the vision language model recognises these behaviours more accurately. Although behaviour recognition on the resampled video produces an excessive number of false “dive” detections, this over-prediction is acceptable. In our application, missing a rare and important behaviour carries a higher penalty than mistakenly labelling an ordinary activity as rare, so a certain level of false positives can be tolerated. In addition, the undersampling routine frames reduce the video’s size from 33.1MB to 9.68MB, illustrating the efficiency of our activity-recognition pipeline.

5 Conclusion

This study introduced a zero-shot framework that couples egocentric video with inertial sensor data to automate the discovery and annotation of animal behaviours. By fusing CLIP-based visual embeddings with HARNet sensor features and directing a VLM to densely caption anomaly-guided segments, our method (i) eliminates the need for handcrafted labels, (ii) identifies previously unseen activities, and (iii) reduces storage requirements through informed frame sampling. A preliminary evaluation on seabird footage demonstrated that resampling guided by IMU-derived anomaly scores yields more accurate behavioural labels than uniform sampling while simultaneously shrinking video size. We are going to undertake a comprehensive evaluation that encompasses the terrestrial brown-bear dataset already collected, as well as additional species and habitats.

Acknowledgments

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References

- [1] Moloud Abdar et al. 2024. A Review of Deep Learning for Video Captioning. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024), 1–20. doi:10.1109/TPAMI.2024.3522295
- [2] Danielle D Brown, Roland Kays, Martin Wikelski, Rory Wilson, and A Klimley. 2013. Observing the unwatchable through acceleration logging of animal behavior. *Animal Biotelemetry* 1 (2013), 20. Issue 1. doi:10.1186/2050-3385-1-20
- [3] Xuguang Duan et al. 2018. Weakly supervised dense event captioning in videos. *Advances in Neural Information Processing Systems* 31 (2018).
- [4] Gaspard Dussert et al. 2024. Zero-shot animal behaviour classification with vision-language foundation models. *Methods in Ecology and Evolution* (2024).
- [5] Zalan Fabian et al. 2023. Multimodal Foundation Models for Zero-shot Animal Species Recognition in Camera Trap Images. *CoRR* (2023).
- [6] Aran Garrod et al. 2021. Video and acceleration records of streaked shearwaters allows detection of two foraging behaviours associated with large marine predators. *PLOS ONE* 16 (7 2021), e0254454. Issue 7. doi:10.1371/journal.pone.0254454
- [7] Benjamin Hoffman et al. 2024. A benchmark for computational analysis of animal behavior, using animal-borne tags. *Movement Ecology* 12, 1 (2024), 78.
- [8] Saiful Islam et al. 2021. Exploring video captioning techniques: A comprehensive survey on deep learning methods. *SN Computer Science* 2, 2 (2021), 1–28.
- [9] Ranjay Krishna et al. 2017. Dense-captioning events in videos. In *Proceedings of the IEEE international conference on computer vision*. 706–715.
- [10] Fei Tony Liu et al. 2008. Isolation forest. In *2008 eighth IEEE international conference on data mining*. IEEE, 413–422.
- [11] Alec Radford et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PMLR, 8748–8763.
- [12] Shuhuai Ren et al. 2024. Timechat: A time-sensitive multimodal large language model for long video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14313–14323.
- [13] Christian Rutz et al. 2007. Video cameras on wild birds. *Science* 318 (11 2007), 765–765. Issue 5851. doi:10.1126/science.1146788
- [14] Kei Tanigaki et al. 2024. Automatic recording of rare behaviors of wild animals using video bio-loggers with on-board light-weight outlier detector. *PNAS nexus* 3, 1 (2024), pgad447.
- [15] Laurens Van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. *Journal of machine learning research* 9, 11 (2008).
- [16] Beth L. Volpov et al. 2015. Identification of prey captures in Australian fur seals (*Arctocephalus pusillus doriferus*) using head-mounted accelerometers: Field validation with animal-borne video cameras. *PLOS ONE* 10 (6 2015), e0128789. Issue 6. doi:10.1371/journal.pone.0128789
- [17] Antoine Yang et al. 2023. Vid2seq: Large-scale pretraining of a visual language model for dense video captioning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10714–10726.
- [18] Hang Yuan et al. 2024. Self-supervised learning for human activity recognition using 700,000 person-days of wearable data. *NPJ digital medicine* 7, 1 (2024), 91.
- [19] Wanrong Zhu et al. 2022. End-to-end Dense Video Captioning as Sequence Generation. In *Proceedings of the 29th International Conference on Computational Linguistics*. 5651–5665.

¹<https://www.youtube.com/watch?v=q5O5tD7f-Rl>